



Cambridge IGCSE™

DESIGN & TECHNOLOGY

0445/11

Paper 1 Product Design

May/June 2026

1 hour 15 minutes



You must answer on the two pre-printed A3 answer sheets.

You will need: Two A3 pre-printed answer sheets (enclosed)
Standard drawing equipment
Coloured pencils

INSTRUCTIONS

- Answer **one** question.
- Use an HB pencil for any drawings and a black or dark blue pen for any writing.
- Write your name, centre number and candidate number in the space on **both** pre-printed answer sheets.
- Answer in the space provided on the answer sheets.
- Do **not** use an erasable pen, staples, paper clips, glue, or correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.
- You may use standard drawing equipment, including coloured pencils.
- At the end of the examination, hand in your named A3 answer sheets. Do **not** fasten them together and do **not** punch holes in the sheets or tie with string.

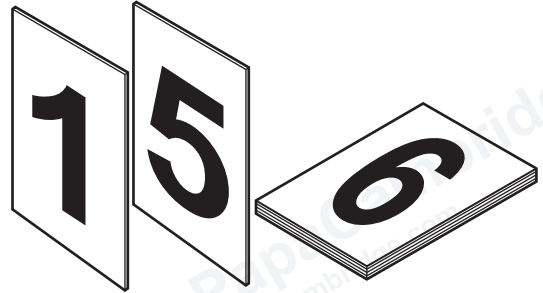
INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].
- All dimensions are in millimetres unless otherwise stated.

This document has 4 pages.

Answer **one** question only on the A3 pre-printed answer sheets provided.

- 1 Team sports are often played on large pitches.

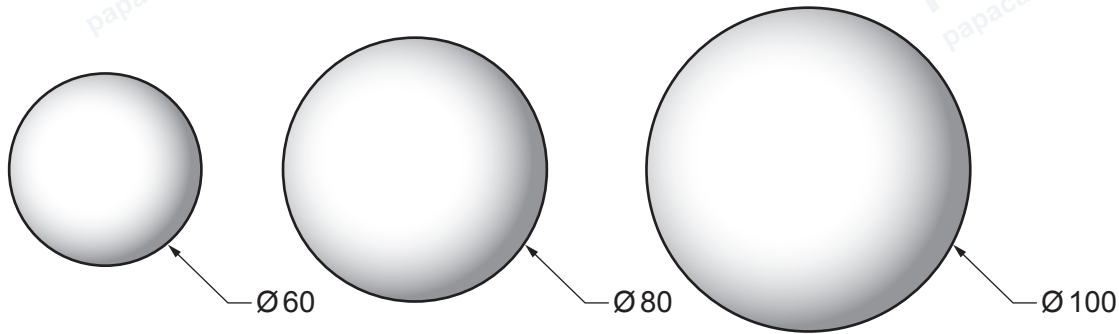


Number boards
400 × 300 × 6
plywood

Design a product to display the scores of two teams, using the number boards shown. The product should be portable and allow the scores to be regularly updated.

- (a) List **four** additional points about the function of such a product that you consider to be important. [4]
- (b) Use sketches and notes to show **two** methods for temporarily displaying items. [4]
- (c) Develop and sketch **three** separate ideas for the product. [12]
- (d) Evaluate your three ideas. Choose **one** idea to develop further and justify your choice. [8]
- (e) Draw, using a method of your own choice, a full solution to the design problem. Include construction details and important dimensions. [12]
- (f) Suggest **two** suitable specific materials for the solution you have drawn in 1(e) and give reasons for your choice. [4]
- (g) Outline a method that could be used to manufacture **one** part of your solution drawn in 1(e). Include the names of the tools used. [6]

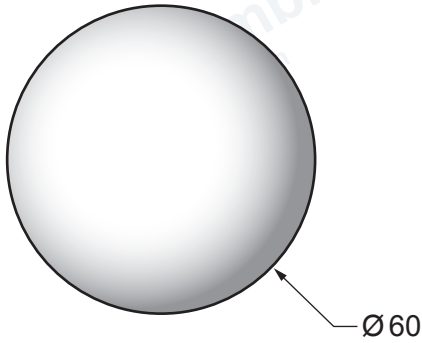
- 2 A sports company wishes to sell a set of three balls for an outdoor game.



Design a piece of packaging to securely hold the three different sized balls shown. The packaging should be able to be stacked on a shelf and allow the balls to be seen.

- (a) List **four** additional points about the function of such packaging that you consider to be important. [4]
- (b) Use sketches and notes to show **two** methods of holding a ball in place only using lightweight graphic materials. [4]
- (c) Develop and sketch **three** separate ideas for the packaging. [12]
- (d) Evaluate your three ideas. Choose **one** idea to develop further and justify your choice. [8]
- (e) Draw, using a method of your own choice, a full solution to the design problem. Include construction details and important dimensions. [12]
- (f) Suggest **two** suitable specific materials for the solution you have drawn in 2(e) and give reasons for your choice. [4]
- (g) Outline a method that could be used to manufacture **one** part of your solution drawn in 2(e). Include the names of the tools used. [6]

- 3 A sports company wishes to develop an outdoor target game.



Design a device to safely propel the foam ball shown at a target. The device should be handheld and use a trigger to release the ball at the target.

- (a) List **four** additional points about the function of such a device that you consider to be important. [4]
- (b) Use sketches and notes to show **two** mechanical methods for propelling an object in a linear direction. [4]
- (c) Develop and sketch **three** separate ideas for the device. [12]
- (d) Evaluate your three ideas. Choose **one** idea to develop further and justify your choice. [8]
- (e) Draw, using a method of your own choice, a full solution to the design problem. Include construction details and important dimensions. [12]
- (f) Suggest **two** suitable specific materials for the solution you have drawn in 3(e) and give reasons for your choice. [4]
- (g) Outline a method that could be used to manufacture **one** part of your solution drawn in 3(e). Include the names of the tools used. [6]

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